OPTIMIZED HIGHWAY NETWORK DESIGN USING MST AND INTERCHANGE HUBS

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Introduction

- Transportation networks must efficiently connect cities while minimizing cost and traveltime.
- This project models cities as nodes in a graphand highways as edges.
- The goal is to design an optimized highway network using algorithmic techniques.

Problem Understanding

- We are given N cities located at coordinates(x, y).
- Each city has a traffic weight Wi representing its importance.
- We can build highways between cities and optionally create interchanges.
- The objective is to minimize total system cost while ensuring all cities remain connected.

Input Specifications

X – Cost to build one interchange

- Y – Cost per kilometer of highway
- N – Number of cities
- City list containing coordinates (x, y) and traffic weight Wi

Output Specifications

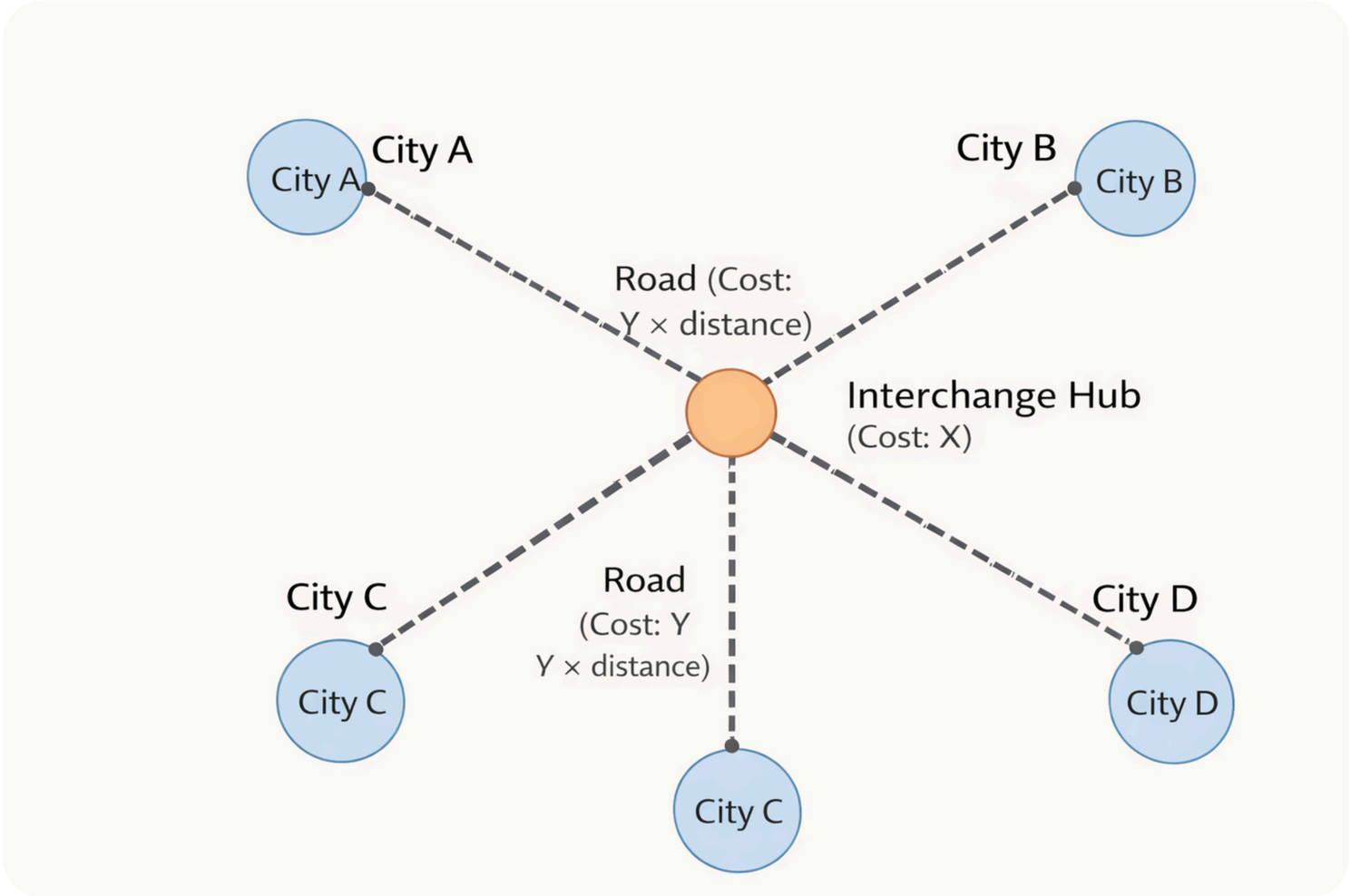
- Coordinates of newly created interchanges
- List of connected highway segments
- Total construction cost (roads +interchanges)

Proposed Algorithmic Approach

Model cities as nodes in a graph withEuclidean distances.

- Step 1: Compute pairwise distances betweencities.
- Step 2: Build Minimum Spanning Tree (MST)using Kruskal’s algorithm to connect cities cheaply.
- Step 3: Compute a Weighted GeometricMedian to act as a potential interchange hub.
- Step 4: Compare two strategies:
  Direct MST connections
  Star network using interchange hub
- Step 5: Choose the network with lower total system cost.

Example Highway Network Diagram



Minimum Spanning Tree Concept

- MST connects all nodes with minimum total edge weight.
- Kruskal’s algorithm is used to construct theMST.
- It ensures minimal road construction costwhile keeping the graph connected.

Interchange Hub Strategy

- A central interchange may reduce total traveldistance.
- The hub location is determined using a weighted geometric median.
- The algorithm compares MST cost with hub-based network cost to choose the best design.

Design Justification

- MST ensures minimum road construction costto connect all cities.
- Weighted geometric median minimizes totalweighted travel distance.
- Interchanges act like Steiner points to shortenoverall paths.
- Comparing both strategies guaranteescost-efficient infrastructure.
- This hybrid design balances:
  - Construction cost
  - Travel efficiency
  - Traffic importance.

Algorithmic Analysis (Asymptotic)

- Distance Matrix Computation: $O(N^2)$
- Sorting edges for Kruskal: $O(E \log E)$
- MST Construction: $O(E \log V)$
- Geometric Median Iteration (Weiszfeld): $O(N \times \text{iterations})$
- Overall Complexity $\approx O(N^2 \log N)$
- Space Complexity: $O(N^2)$ for distance storage.
- Efficient for moderate city network sizes.

Conclusion

- The proposed approach combines MST and hub optimization.
- It balances construction cost and travelefficiency.
- This method reflects real-world highway planning strategies.
- Future improvements could include multi-hub clustering.

THANK YOU